

Applications of Graph Theory — Solutions

Problem 1. (MEMO 2011 Individual Problem 2) Let m and n be positive integers. Some squares of an $m \times n$ board are coloured red. A sequence $a_1, a_2, \dots, a_{2r}$ of $2r \geq 4$ pairwise distinct red squares is called a *bishop circuit* if for every $k \in \{1, \dots, 2r\}$, the squares a_k and a_{k+1} lie on a diagonal, but the squares a_k and a_{k+2} do not lie on a diagonal (here $a_{2r+1} = a_1$ and $a_{2r+2} = a_2$).

In terms of m and n , determine the maximum possible number of red squares on an $m \times n$ board without a bishop circuit.

(Remark. Two squares lie on a diagonal if the line passing through their centres intersects the sides of the board at an angle of 45° .)

Solution. The maximum value is $2m + 2n - 4$ if $m, n \geq 2$, is n if $m = 1$, and is m if $n = 1$.

The cases $m = 1$ or $n = 1$ are trivial since we can colour all squares red. Suppose $m, n \geq 2$. Consider a graph G whose vertices are the diagonals of the board such that two vertices are joined by an edge if the corresponding diagonals intersect at a red square. Note that a bishop circuit corresponds to a cycle in G . Therefore, G has no cycle.

Also, by considering the chessboard colouring, we see that each diagonal only covers squares of the same colour, and so diagonals of different colours cannot intersect. This shows G has at least two connected components.

The number of vertices in G is the number of diagonals, which is $2m + 2n - 2$. It follows that the number of edges is at most $2m + 2n - 4$. Note that the number of edges is exactly the number of red squares.

It remains to provide a construction with $2m + 2n - 4$ red squares. Indeed, we can colour every square in the first column, the last column, and the last two rows red. Consider the chessboard colouring. It is not hard to see that all red+black squares can be travelled to one another (via diagonal moves). Also, every black diagonal contains at least one red square (on the left, on the right or at the bottom). This implies all vertices corresponding to the black diagonals are connected. The same holds for white diagonals, and so G only contains two connected components. As the number of edges is two less than the number of vertices, both components are trees as desired. $\square$

Problem 2. (APMO 2013 Problem 4) Let a and b be positive integers, and let A and B be finite sets of integers satisfying

- (i) A and B are disjoint;
- (ii) if an integer i belongs to either to A or to B , then either $i + a$ belongs to A or $i - b$ belongs to B .

Prove that $a|A| = b|B|$. (Here $|X|$ denotes the number of elements in the set X .)

Solution. Consider a directed graph G whose vertices are the elements in $A \cup B$. For each $i \in (A \cup B)$, we draw the edge from i to $i + a$ (if $i + a \in A$) and draw the edge from i to $i - b$ (if $i - b \in B$). By (ii), the out-degree of each vertex is at least 1. Also, it is obvious that the in-degree of each vertex is at most 1. As the sum of in-degrees is equal to the sum of out-degrees, each vertex must have in-degree 1 and out-degree 1. Thus, G is the union of some disjoint cycles.

In each cycle, suppose there are x edges of the form $i \rightarrow i + a$ and y edges of the form $i \rightarrow i - b$. Then we have $xa = yb$ since the number is unchanged after travelling for one cycle. This means the ratio of the number of elements in A (which is x) to the number of elements in B (which is y) in this cycle is $b : a$. Since this holds for every cycle, we have $|A| : |B| = b : a$. $\square$

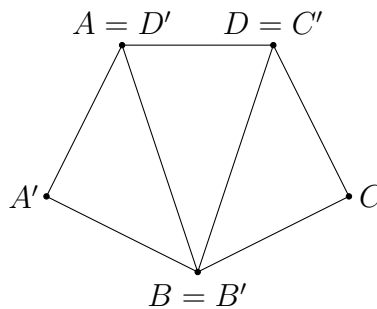
Problem 3. (Kharkiv Masters Tournament 2019 Grade 10–11 Problem 6) Consider any triangulation T of a convex polygon $\mathcal{P}$ with 100 sides. For each diagonal AC of $\mathcal{P}$, we say that it is *good* if there exist two more vertices B and D of $\mathcal{P}$ such that $ABCD$ is convex, and all the segments AB, BC, CD, DA, BD are drawn in T . Find the maximum number of pairs of good diagonals which intersect inside $\mathcal{P}$.

Solution. The maximum value is 144.

Consider a graph G whose vertices are the triangles in T such that two vertices are joined by an edge if the corresponding triangles share a common edge. Note that G is a tree since otherwise the cycle in G must enclose a vertex inside, which is impossible since $\mathcal{P}$ is convex.

For any good diagonal AC , the vertices B and D given in the question are uniquely determined. Indeed, suppose both B, D and B', D' satisfy the conditions where $ABCD$ and $AB'CD'$ have the same orientation. WLOG assume $ABB'CD$ is convex. Then the diagonals $B'D'$ and BC must cross each other, which is impossible as T is a triangulation. Thus, there is a bijection between the good diagonals and the edges in G (mapping AC to the edge joining $\triangle ABD$ and $\triangle CBD$).

Now, consider a pair of good diagonals AC and $A'C'$ which intersect inside $\mathcal{P}$. Suppose $ABCD$ and $A'B'C'D'$ are the corresponding quadrilaterals. WLOG assume A' and B lie on the same side of the line AC (so C' and D lie on the other side). Also, we may assume A' lies between A and B . If $B' \neq B$, then B' must lie between A' and B . But then $B'C'$ must cross AB . Thus, we must have $B' = B$.



In order that $B'C'$ does not cross AD and CD , we must have $C' = D$. In order that AB does not cross $C'D'$ and $A'D'$, we must have $D' = A$. Thus, $ABCD$ and $A'B'C'D'$ share a common triangle. So

the edge joining $\triangle BCD$ and $\triangle ABD$ shares a common endpoint with the edge joining $\triangle A'B'D'$ and $\triangle B'C'D'$. Conversely, it is clear that any pair of edges in G sharing a common endpoint corresponds to such a pair of intersecting good diagonals.

Thus, it remains to maximize the number of pairs of edges in G sharing a common endpoint. Let $d_1, d_2, \dots, d_{98}$ be the degrees of the vertices in G . Note that $d_j = 1, 2, 3$ because of the triangulation.

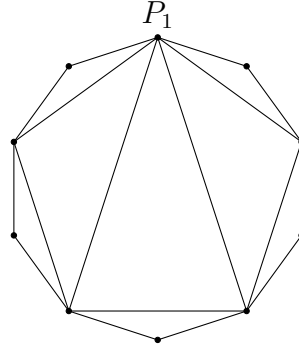
Also, we need to maximize $\sum_{j=1}^{98} \binom{d_j}{2}$. By the majorization theorem, the maximum value of this quantity is attained when all d_j 's are 1 or 3. Since the number of edges of G is 97 (as it is a tree), we have $d_1 + d_2 + \dots + d_{98} = 2(97) = 194$. Thus, it is not hard to see that 48 of the d_j 's are equal to 3, and 50 of them are equal to 1. So the maximum value is

$$48 \binom{3}{2} + 50 \binom{1}{2} = 144.$$

One example of the triangulation of $P_1 P_2 \dots P_{100}$ is

$$\triangle P_1 P_2 P_3, \triangle P_3 P_4 P_5, \dots, \triangle P_{99} P_{100} P_1, \triangle P_1 P_3 P_5, \triangle P_1 P_5 P_7, \dots, \triangle P_1 P_{97} P_{99}.$$

It is not hard to see that each vertex in G has degree 1 or 3, so the maximum value is attained. $\square$



Problem 4. (De Bruijn sequence) Let n be a positive integer. Prove that there exists a binary string $a_1 a_2 \dots a_{2^n}$ of length 2^n (where $a_j = 0, 1$ for each j) such that for every binary string $b_1 b_2 \dots b_n$ of length n (where $b_j = 0, 1$ for each j), there exists k such that $a_{k-1+j} = b_j$ for $j = 1, 2, \dots, n$ (where the indices of a_i are taken modulo 2^n).

Solution. Consider a directed graph G whose vertices are all the binary strings of length $n - 1$. We draw an edge from each vertex $a_1 a_2 \dots a_{n-1}$ to the vertex $a_2 a_3 \dots a_{n-1} a_n$ where $a_n = 0, 1$. Also, we label this edge by $a_1 a_2 \dots a_n$.

Since the in-degree and out-degree of each vertex are the same (both are equal to 2), and G is strongly connected (to go from $a_1 a_2 \dots a_{n-1}$ to $b_1 b_2 \dots b_{n-1}$, we just follow the edges $a_1 a_2 \dots a_{n-1} b_1$,

$a_2a_3 \cdots a_{n-1}b_1b_2, \dots, a_{n-1}b_1b_2 \cdots b_{n-1}$), we know that G has an Eulerian circuit. Suppose the edges in this Eulerian circuit are

$$a_1a_2 \cdots a_n, a_2a_3 \cdots a_{n+1}, \dots, a_{2^n}a_1a_2 \cdots a_{n-1}$$

in order. Then the binary string $a_1a_2 \cdots a_{2^n}$ satisfies the condition because each edge represents a distinct binary string of length n . $\square$

Problem 5. (Serbia TST 2021 Problem 4) Given that $a_1, a_2, \dots, a_{2020}$ are integers, find the maximal number of subsequences $a_i, a_{i+1}, \dots, a_j$ ($0 < i \leq j < 2021$) with sum 2021.

Solution. The maximum value is 1021110.

Let $b_0 = 0$ and $b_k = a_1 + a_2 + \cdots + a_k$. Consider a graph G with vertices $0, 1, \dots, 2020$ such that vertices i and j are joined by an edge if $b_i - b_j = \pm 2021$. Note that if $a_{i+1} + a_{i+2} + \cdots + a_j = 2021$, then $b_j - b_i = 2021$, and so there is an edge joining vertices i and j . Thus, the number of such subsequences cannot exceed the number of edges in G .

If G has an odd cycle $i_1, i_2, \dots, i_k$, then we have

$$0 = (b_{i_1} - b_{i_2}) + (b_{i_2} - b_{i_3}) + \cdots + (b_{i_k} - b_{i_1}) = \pm 2021 \pm 2021 \pm \cdots \pm 2021.$$

This is impossible since k is odd. Therefore, G has no odd cycle, which means it is a bipartite graph. The maximum number of edges in G is the maximum value of $m(2021 - m)$ (with m being a nonnegative integer), which is $1010 \cdot 1011 = 1021110$.

Now, consider the sequence $a_k = 0$ for $k \neq 1010$ and $a_{1010} = 2021$. Then all subsequences $a_i, a_{i+1}, \dots, a_j$ with $1 \leq i \leq 1010 \leq j \leq 2020$ have sum 2021. There are $1010 \cdot 1011 = 1021110$ such subsequences. So the maximum value can be attained. $\square$

Problem 6. (USA TST 2010 Problem 6) Let T be a finite set of positive integers greater than 1. A subset S of T is called *good* if for every $t \in T$ there exists some $s \in S$ with $\gcd(s, t) > 1$. Prove that the number of good subsets of T is odd.

Solution. Consider a graph G whose vertices are the elements in T such that two vertices a and b are joined by an edge if $\gcd(a, b) > 1$. A good subset means a set S of vertices such that each vertex in G is adjacent to at least one vertex in S .

For each set X of vertices, let $f(X)$ be the set of vertices which are not adjacent to any vertex in X . Then X is good if and only if $f(X) = \emptyset$. We say that a pair (X, Y) of sets of vertices is *nice* if the distance between every $x \in X$ and $y \in Y$ is at least 2. Let N be the number of nice pairs.

For each fixed X , there are $2^{|f(X)|}$ nice pairs of the form (X, Y) since Y can be any subset of $f(X)$. This is odd if $f(X) = \emptyset$ and is even otherwise. Therefore, N has the same parity as the number of good subsets. But then notice that (X, Y) is nice if and only if (Y, X) is nice, and the only nice pair (X, X) is $X = \emptyset$. Therefore, N must be odd as desired. $\square$